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# Retinal graph neural network for segmentation of retinal vasculature and foveal avascular zone from OCTA images

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**Abstract**

**Objectives:** Diabetic Retinopathy (DR) causes major vision loss, requiring precise segmentation of retinal vessels and the Foveal Avascular Zone (FAZ). Accurate structural masks enable quantitative biomarkers that support early diagnosis and long-term monitoring.

**Methods:** We propose a Retinal Graph Neural Network (RGNNNet) for OCTA segmentation. It combines multi-scale feature extraction with a graph representation, where node relations derive from an affinity matrix of feature maps. A symmetric normalization strategy stabilizes graph propagation and integrates local–global vascular context. A hybrid Dice–Focal loss refines fine-structure segmentation.

**Results:** On OCTA-500, RGNNNet achieved superior Dice and IoU to existing methods. For FAZ, it attained Dice values of 96.78 % (6 mm) and 98.02 % (3 mm), and maintained 0.915 on ROSE-0 without retraining. It outperformed baselines by 1–3 % Dice for other classes and remained lightweight (0.83 M params, 11.25 ms per 400 × 400 image).

**Conclusions:** By coupling residual feature learning with graph-based relational reasoning, RGNNNet provides accurate structure-specific masks that can serve as a foundation for downstream biomarker extraction. Its compact design and stable generalization highlight its potential for large-scale ophthalmic screening and integration into clinical workflows.

**Keywords:** biomedical image processing; deep learning; diabetic retinopathy; image segmentation; graph neural networks

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## Introduction

Diabetic Retinopathy (DR) is a progressive retinal disease caused by prolonged hyperglycemia, leading to vascular damage in the retina, particularly affecting capillaries, arteries, veins, and the Foveal Avascular Zone (FAZ) [1]. In its early stages, known as Non-Proliferative Diabetic Retinopathy (NPDR), DR is characterized by microaneurysms, capillary non-perfusion, and pericyte loss, all of which contribute to retinal ischemia and edema [2]. If untreated, the disease advances to Proliferative Diabetic Retinopathy (PDR), marked by neovascularization and severe retinal damage that can ultimately result in irreversible vision loss [3]. Structural changes in the FAZ, such as enlargement and irregularities, are particularly significant as they correlate with macular ischemia and the progression of DR [4]. According to surveys, more than 126 million people are affected by DR, and a conservative estimate projects the number of patients to increase to 190 million by 2030, highlighting the urgent need for intervention [5]. Therefore, detecting these changes early is critical to prevent vision-threatening complications. Optical Coherence Tomography Angiography (OCTA) is a preferred imaging modality for DR diagnosis because it provides high-resolution visualization of retinal vasculature, including capillaries, arteries, veins, and the FAZ, without requiring contrast agents [6]. This non-invasive technique identifies early DR biomarkers, such as reduced vessel density and FAZ abnormalities, which are challenging to detect with traditional imaging methods [4]. However, manual analysis of OCTA images is time-intensive and prone to variability among clinicians, particularly when interpreting the complex retinal vascular network [7].

Deep Learning (DL) approaches have been increasingly used to automate OCTA image analysis, offering faster and more reliable segmentation of retinal features compared to manual methods. These models are particularly effective in analyzing intricate vascular patterns caused by DR [7]. Existing approaches often require assistance in segmenting vessels of varying sizes, shapes, and orientations. Capillaries, in particular, pose challenges due to their delicate, mesh-like structure, leading to segmentation errors and fuzzy boundaries in pathological regions [8]. Additional issues, such as

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motion artifacts, blinking-induced distortions, and shadowing caused by vitreous floaters, further degrade the quality of OCTA images and hinder accurate segmentation [9]. Low signal-to-noise ratios and the scarcity of diverse, well-annotated datasets also limit the generalizability of current methods in clinical settings [10, 11]. To overcome these limitations, this study introduces a novel graph-based approach for OCTA image segmentation, leveraging the structural properties of retinal vasculature. Using message-passing mechanisms, the model incorporates contextual information by portraying OCTA images as graphs with nodes that capture vascular characteristics. The segmentation accuracy for capillaries, arteries, veins, and the FAZ is greatly improved by this formulation, which considers the morphological diversity and inter-class differences present in retinal vessels. The suggested approach seeks to enable prompt therapies to preserve vision and enhance patient outcomes by facilitating early and accurate DR diagnosis. In contrast to prior CNN or transformer-based models, our novelty lies in three aspects: (i) dynamic graph construction with symmetric normalization inside the RGNN block, which stabilizes message passing without requiring handcrafted topologies, (ii) integration of residual units at every encoder stage to ensure continuity of vascular features while deepening feature extraction, and (iii) adaptive skip refinement that fuses encoder and decoder features to preserve fine vascular boundaries. This end-to-end formulation enables RGNNNet to simultaneously capture both local vessel continuity and global contextual dependencies, a capability that has not been achieved in previous OCTA segmentation frameworks.

## Related work

Medical image segmentation has advanced significantly from traditional algorithms to statistical models and techniques employing DL. Early approaches to DR segmentation highly rely on manual annotations along with handcrafted features, which require a significant amount of time and introduce observer variability. The advent of DL led to a revolution in the segmentation of medical images due to hierarchical feature extraction and representation learning, thus performing automatic and accurate analysis on retinal structures [1]. Despite these benefits, poor generalization toward unseen datasets, difficulty in capturing subtle features in OCTA images, and low robustness in pathological scenarios need improvement, thus leading to the development of more complex architectures. DL models have successfully segmented retinal structures, particularly in fundus images and OCTA scans. U-Net [12] has been a foundational architecture, with its encoder-decoder structure and skip connections effectively preserving spatial information. Nonetheless, U-Net struggles to capture fine-grained details, such as thin capillaries. Enhanced variants, such as Attention U-Net [13], UNet++ [14], and UNet 3+ [15], introduced mechanisms like attention, nested skip connections, and full-scale feature aggregation to address these shortcomings. More recently, transformer-based architectures, such as TransUNet [16] and UNETR [17], have incorporated global self-attention and robust feature fusion, achieving notable improvements in segmentation. However, these models still face challenges in segmenting intricate structures, such as non-perfusion areas, and differentiating retinal vessel types.

Further optimizations have been observed in architectures such as CE-Net [18] and R2U-net [19], which utilize context encoders and recurrent units to enhance feature extraction and fusion, particularly in complex retinal scenarios. Task-specific models, such as AV-Net [20] and CS-Net [21], have focused on DR-specific requirements, including the ability to distinguish between arteries, veins, and capillaries in OCTA images. Despite their advancements, these models often fail to delineate non-perfusion areas and capture the long-range dependencies essential for accurate DR diagnosis. Domain adaptation and semi-supervised learning techniques have also been explored to improve segmentation performance on diverse datasets. Semi-supervised frameworks leverage labeled and unlabeled data to extract meaningful features and reduce dependency on labor-intensive annotations [22]. Adversarial domain adaptation has been utilized to address variability in OCTA imaging protocols and hardware, enhancing generalization across datasets [23]. Despite these advancements, reliably segmenting non-perfusion areas and accurately delineating vascular boundaries in pathologically altered retinas remains challenging. OCTA-Net [11], one of the early OCTA-based segmentation models, utilized dual U-Net architectures but encountered difficulties in segmenting thin capillaries and non-perfusion regions. Subsequent frameworks, such as IPN V2 and IPN V2+ [24], improved FAZ segmentation through 3D-to-2D projections yet struggled with irregular boundaries and overlapping structures. Modules such as distance-aware joint learning and boundary regression have shown potential in addressing these challenges, although they are limited in modeling long-range dependencies [8, 25]. Task-specific advancements, such as the IPN V2 framework's distance-aware joint learning and boundary regression modules, have demonstrated potential in addressing issues like blurred boundaries and irregular vessel overlaps [24]. The availability of detailed annotations from OCTA-based datasets, such as ROSE [11] and OCTA-500 [26], has further

enhanced model training, enabling the accurate segmentation of both capillaries and larger vessels. However, the lack of mechanisms to capture long-range dependencies and contextual relationships underscores the need for hybrid architectures.

The segmentation of retinal vasculature and the FAZ from OCTA images has become increasingly important for the early diagnosis of DR. Conventional deep learning methods, particularly convolutional neural networks (CNNs), have achieved notable results in medical imaging. However, their limited ability to capture non-local dependencies and structural relationships within the retina often constrains their performance in segmenting complex vascular regions. Recent advances have explored the potential of Graph Neural Networks (GNNs) to overcome these limitations. Graph-based methods have recently introduced a new paradigm for medical image segmentation, modeling images as graphs to capture local and global contextual relationships. Graph Neural Networks (GNNs), which approach segmentation as a node classification task, have effectively leveraged graph structures to represent interdependencies [27]. Although GNNs are relatively new to OCTA segmentation, their ability to classify vascular structures, such as arteries and veins, using message-passing algorithms and topological constraints has shown promise. Hybrid models that combine CNNs and GNNs have enhanced segmentation accuracy for capillaries, arteries, veins, and the FAZ, particularly in addressing non-perfusion areas and overlapping structures [28]. A study focusing on fundus images proposed a GCN-based classification approach that extracted topological features using a variational autoencoder. This method demonstrated enhanced accuracy and sensitivity in DR stage classification, emphasizing the advantages of integrating graph-based representations with deep learning for retinal analysis [29]. Building on this, researchers introduced a vision-GNN pipeline tailored for OCT imaging, applying graph-based modeling for multi-class classification of retinal diseases. Their framework successfully encoded spatial relationships within OCT scans, achieving a classification accuracy of over 99 % and establishing the relevance of GNNs in OCT-based diagnostics [30]. Further progress was made by hybrid architectures that combine GNNs with attention mechanisms. One such model, graph neural network-guided vision transformer enhanced network (G2ViT), unified CNNs, GNNs, and vision transformers to capture both local textures and global structural patterns in vessel segmentation. This framework incorporated edge-aware modules and multi-level fusion, significantly improving the segmentation of fine vessel networks in both retinal and coronary angiography datasets [31]. In another direction, a hybrid graph-based learning strategy was proposed for detecting carotid artery stenosis using OCTA images. This approach helps to distinguish arterial and venous features to build a composite graph, demonstrating that vascular-specific representations improve classification accuracy in vascular-related pathologies. Although not directly targeting DR, this method validated OCTA's capability in supporting graph-based analysis for microvascular abnormalities [32]. These studies collectively highlight the growing effectiveness of GNNs in retinal image analysis. They reveal that GNNs not only model spatial hierarchies more effectively than CNNs but also provide improved generalization across anatomical variations. However, challenges remain, particularly in integrating multi-scale features, preserving boundary precision, and delineating the FAZ.

Unlike prior GNN-based retinal segmentation frameworks, which often focus on classification tasks or rely on handcrafted graph structures [29, 30], the proposed RGNNNet introduces a novel integration of a Retinal Graph Neural Network (RGNN) module within a residual encoder-decoder architecture, specifically tailored for OCTA-based pixel-level segmentation. The model dynamically constructs graph representations via learned feature affinities and applies symmetric normalization for stable message propagation, avoiding the need for predefined topologies [32]. This capability enables the effective modeling of contextual dependencies critical for delineating fine vascular structures and pathological regions, such as non-perfusion zones. Additionally, the incorporation of adaptive skip refinement blocks facilitates accurate boundary preservation and high-resolution reconstruction across all anatomical targets.

Recent studies have highlighted the value of graph reasoning in retinal analysis. On fundus images, G2ViT couples graph guidance with a transformer backbone, improving connectivity in thin vessels [31]. VGA-Net and GViT-RSNet add graph attentional modules that emphasize topology and small structures [33, 34]. In OCTA, Eye-AD builds a multilevel graph across retinal layers and reports stable gains on a large multicenter cohort [35]. Clinically focused OCTA work complements this direction, with a widefield network segmenting nonperfusion areas across multiple plexuses and showing strong agreement with expert labels [36]. FAZ segmentation across multiple diseases has also advanced, with a distilled transformer that generalizes beyond diabetic retinopathy [37]. Motivated by these trends, we learn feature affinities with symmetric normalization inside a compact RGNN block and integrate it with a straightforward encoder and decoder.

Taken together, these studies clarify the remaining gap: dynamic graph construction with explicit symmetric normalization and end-to-end pixelwise learning for *multi-*

*structure* OCTA segmentation are still underexplored. In contrast to transformer-guided graph cues in fundus imaging [31] and graph-attentional add-ons [33, 34], our RGNN block performs message passing over a *learned*, symmetrically normalized affinity that is computed directly from OCTA features and optimized jointly with the network. Unlike multilevel graph analysis aimed at clinical classification [35], and task-specific frameworks focusing on widefield nonperfusion delineation [36] or FAZ-only segmentation with distillation [37], our formulation targets concurrent, pixel-accurate masks for capillaries, arteries, veins, and FAZ. This positioning motivates our unified design, which couples residual encoding with symmetric-normalization graph reasoning and adaptive skip refinement to capture both local vessel continuity and global vascular context within a single architecture. In contrast to existing methods such as IPN-V2 [24] and G2ViT [31], which either focus on boundary regression or require external attention modules, RGNNNet achieves end-to-end segmentation by unifying residual learning, multi-scale feature fusion, and graph-based relational modeling. As a result, the model consistently achieves superior performance across capillaries, arteries, veins, and the FAZ, validating its novelty and efficacy in addressing the challenges inherent to OCTA segmentation.

# Methodology

In this section, the architecture and components of the proposed RGNNNet model are detailed, with an emphasis on the contributions of each module to segmentation performance on OCTA images. The key design choices and technical innovations are outlined to demonstrate the approach's effectiveness. Figure 1 illustrates the overall architecture of the RGNNNet, which integrates a Residual Unit structure with the RGNN block. This design enables accurate segmentation of retinal vasculature and the FAZ, highlighting the novelty and utility of the proposed framework. Throughout this paper, **RGNNNet** refers to the complete segmentation architecture with the encoder, the **RGNN block**, and the decoder (Figure 1). The **RGNN block** refers only to the graph reasoning module at the bottleneck (Section "Proposed retinal graph neural network block for retinal vasculature and FAZ segmentation"; Figure 2). We use RGNNNet when discussing end-to-end training, inference, and final masks. We use the RGNN block when describing graph construction, affinity normalization, and message passing inside the module. We denote the image height and width by $H$ and $W$, respectively. The number of input feature channels is $C_{\text{in}}$ and the channel dimension used by the graph module is $C_{\text{graph}}$. A mini-batch has size $B$. Accordingly, the encoder input has shape $\mathbf{X} \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$, and the bottleneck feature map that feeds the RGNN block has shape $\mathbf{Z} \in \mathbb{R}^{B\times C_{\text{graph}}\times H'\times W'}$.

## Encoder

The encoder extracts multi-scale features critical for capturing and segmenting capillaries, veins, arteries, and the FAZ in OCTA images. Each stage employs residual blocks with increasing filter sizes, refining spatial and structural representations for detailed segmentation. Each stage of the encoder employs two residual units arranged sequentially to form a stacked residual block. This configuration ensures progressive refinement of spatial and contextual features by preserving continuity across convolutional layers. The sequential design is particularly effective for OCTA segmentation tasks, where accurate delineation of arteries, veins, capillaries, and FAZ requires both depth-wise feature encoding and preservation of vascular continuity. By avoiding early fusion or concatenation, the encoder maintains a clear flow of structural information, which is essential for learning both coarse and fine retinal patterns. The encoder consists of five sequential stages designed to progressively extract deep hierarchical features from OCTA images. Each stage comprises two residual blocks, Residual Unit A and Residual Unit B, which are applied in sequence to enhance feature refinement while preserving spatial continuity. Let $\mathbf{x}_0 \in \mathbb{R}^{B\times 1\times H\times W}$ denote the input OCTA image. The operators $\mathscr{F}_A(\cdot)$ and $\mathscr{F}_B(\cdot)$ represent the outputs of Residual Units A and B, respectively, while $P(\cdot)$ denotes a $2 \times 2$ max-pooling operation for spatial downsampling The encoder uses residual blocks at each stage; each stage contains two Residual Units, and the resulting features are passed to the RGNN block at the bottleneck. This arrangement refines features at the same spatial resolution before downsampling, which improves gradient flow, stabilizes optimization, and helps preserve thin vessel continuity. This design was empirically validated: when only one Residual Unit per stage was used, segmentation Dice and IoU scores were consistently lower across all structures. Thus, adopting two Residual Units per stage provides a stable improvement in performance without altering the overall training pipeline.

**Stage 1:** The input $\mathbf{x}_0$ is processed by two residual blocks with 16 filters, focusing on fine-grained local patterns such as capillaries:

$$\mathbf{y}_1 = \mathscr{F}_B(\mathscr{F}_A(\mathbf{x}_0)) \tag{1}$$

**Stage 2:** The output $\mathbf{y}_1$ is downsampled using a $2 \times 2$ max-pooling operation and passed through two residual

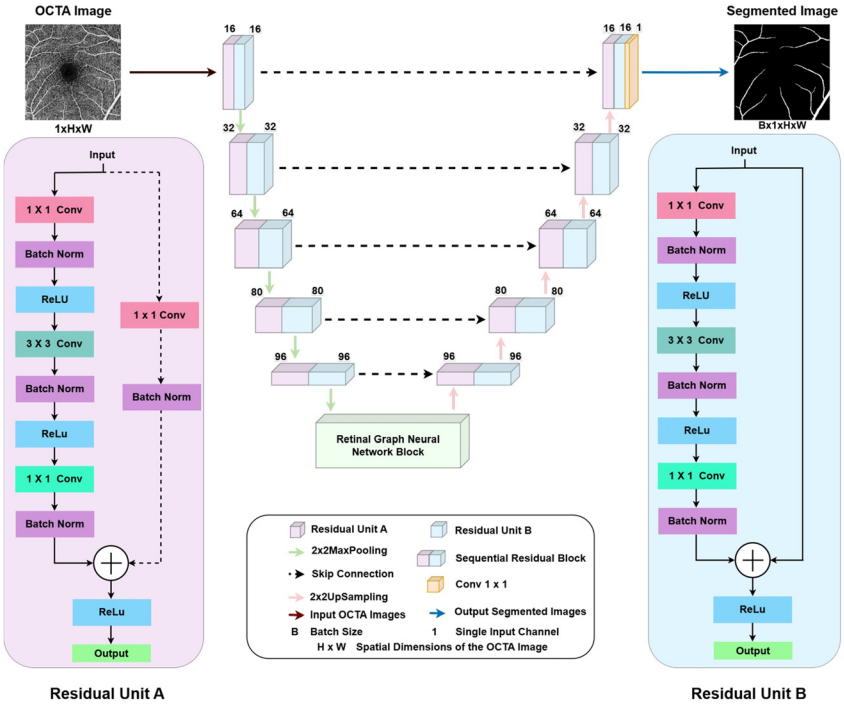



**Figure 1:** RGNNNet overview: pipeline with encoder, RGNN block, and decoder for OCTA segmentation.

blocks with 32 filters to capture intermediate-level vessel structures:

$$\mathbf{y}_2 = \mathcal{F}_B(\mathcal{F}_A(P(\mathbf{y}_1))) \tag{2}$$

**Stage 3:** Residual blocks with 64 filters are applied to capture complex branching patterns and vessel bifurcations:

$$\mathbf{y}_3 = \mathcal{F}_B(\mathcal{F}_A(P(\mathbf{y}_2))) \tag{3}$$

**Stage 4:** This stage uses 80 filters to enhance representations of the foveal avascular zone (FAZ) and larger arterial branches:

$$\mathbf{y}_4 = \mathcal{F}_B(\mathcal{F}_A(P(\mathbf{y}_3))) \tag{4}$$

**Stage 5:** Finally, 96 filters are employed to learn high-level vascular context and relationships:

$$\mathbf{y}_5 = \mathcal{F}_B(\mathcal{F}_A(P(\mathbf{y}_4))) \tag{5}$$

The hierarchically encoded features $\mathbf{y}_1$ through $\mathbf{y}_5$ integrate multi-scale vascular patterns and are passed to the RGNN block for graph-based contextual reasoning. This multi-stage encoder efficiently captures retinal structures across multiple spatial scales. By hierarchically integrating capillaries, veins, arteries, and FAZ-specific features, it lays a strong foundation for robust downstream segmentation.

## Proposed retinal graph neural network block for retinal vasculature and FAZ segmentation

The RGNN block, as illustrated in Figure 2, models OCTA images as graph structures to capture the relationships between retinal regions, enabling accurate segmentation of capillaries, veins, arteries, and the FAZ. Each pixel or region in the OCTA image is treated as a graph node, while the edges represent distances between nodes, capturing both local and global structural dependencies.

### Input and initial feature transformations

The input to the RGNN block is a feature map $\mathbf{I} \in \mathbb{R}^{W \times H \times C_{\text{in}}}$. The feature map undergoes three parallel $1 \times 1$ convolution operations, producing transformed features:

$$\mathbf{U}_1 = \text{Conv}_1(\mathbf{I}), \quad \mathbf{U}_2 = \text{Conv}_2(\mathbf{I}), \quad \mathbf{U}_3 = \text{Conv}_3(\mathbf{I}), \tag{6}$$

where $\text{Conv}_1$, $\text{Conv}_2$, $\text{Conv}_3$ are $1 \times 1$ convolutional layers, reducing the feature dimension to $C_{\text{graph}}$. The dimension $C_{\text{graph}}$ represents the reduced feature space after the $1 \times 1$ convolutions, which is specifically designed to capture relevant graph-based representations of the retinal structures for further processing.

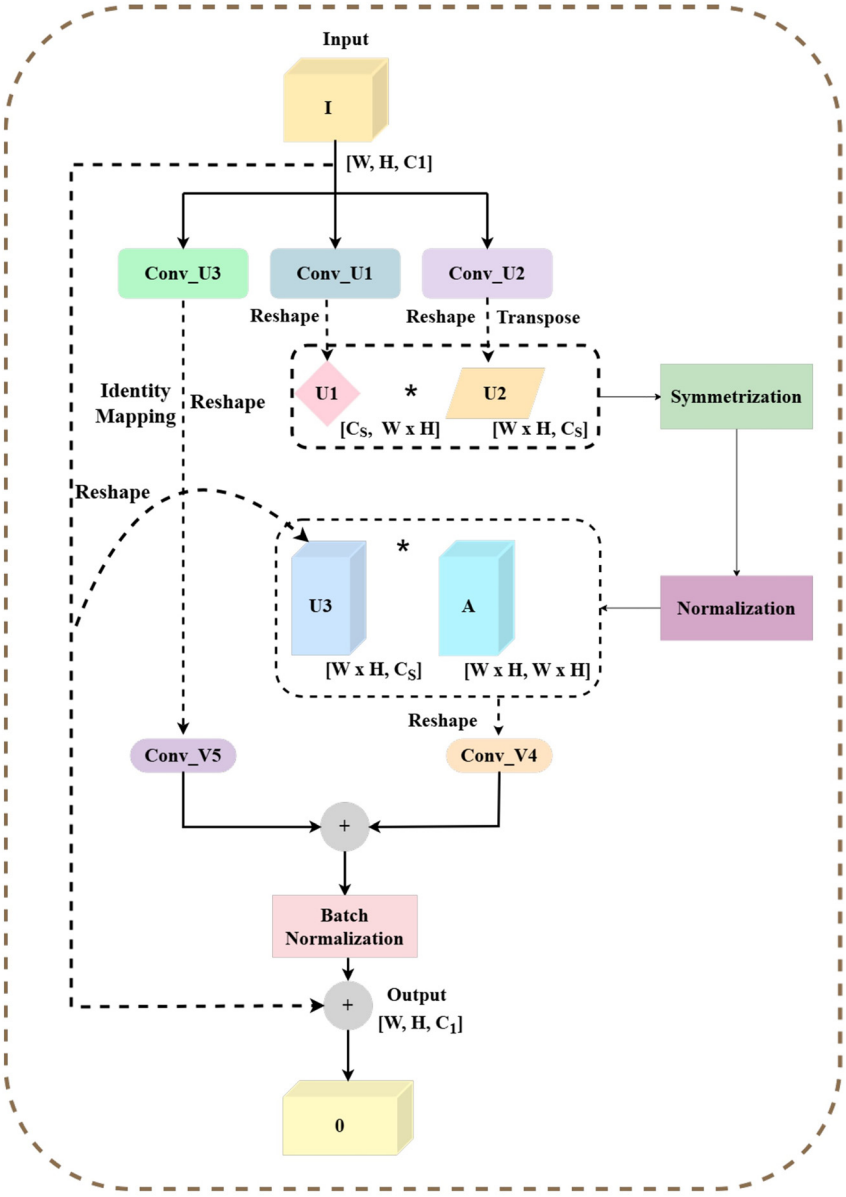



**Figure 2:** Proposed RGNN block (module inside RGNNNet): graph construction, symmetric normalization, and propagation.

### Affinity matrix construction

To represent node-to-node relationships in the graph, the feature maps $\mathbf{U}_1$ and $\mathbf{U}_2$ are reshaped into matrices of size $C_{\text{graph}} \times (W \times H)$ and $(W \times H) \times C_{\text{graph}}$, respectively. Their product generates the affinity matrix $\mathbf{A} \in \mathbb{R}^{(W\times H)\times(W\times H)}$:

$$\mathbf{A} = \mathbf{U}_1^{\top}\mathbf{U}_2. \tag{7}$$

This matrix captures the relationships between pixels (nodes), where higher values in $\mathbf{A}$ signify stronger connections. To ensure bidirectional symmetry, the affinity matrix is symmetrized:

$$\mathbf{A}_{\text{sym}} = \frac{\mathbf{A} + \mathbf{A}^{\top}}{2}. \tag{8}$$

Normalization is applied to stabilize the graph representation:

$$\mathbf{A}_{\text{norm}} = \mathbf{D}^{-\frac{1}{2}}\mathbf{A}_{\text{sym}}\mathbf{D}^{-\frac{1}{2}}, \tag{9}$$

where $\mathbf{D}$ is the degree matrix of $\mathbf{A}_{\text{sym}}$.

### Graph propagation and feature enhancement

Each node in the graph corresponds to a region in the OCTA image, which could belong to distinct vascular structures

(e.g., capillaries, veins, arteries, or the FAZ). The normalized affinity matrix $\mathbf{A}_{\text{norm}}$ is used to propagate information across nodes. The reshaped feature map $\mathbf{U}_3 \in \mathbb{R}^{C_{\text{graph}} \times (W \times H)}$ interacts with $\mathbf{A}_{\text{norm}}$, enhancing node features based on their neighbors:

$$\mathbf{U}_{\text{enh}} = \mathbf{A}_{\text{norm}} \mathbf{U}_3. \tag{10}$$

The enhanced features are reshaped back to $\mathbb{R}^{W \times H \times C_{\text{graph}}}$ and processed through a $1 \times 1$ convolution:

$$\mathbf{F}_{\text{graph}} = \text{Conv}_4(\mathbf{U}_{\text{enh}}). \tag{11}$$

#### Skip connection and output

A skip connection combines the original input $\mathbf{I}$ and the output of a parallel convolution $\text{Conv}_5$ applied to $\mathbf{U}_3$. $\text{Conv}_5$ in the skip connection uses a $1 \times 1$ kernel size. This ensures feature dimensions are aligned when adding the outputs.

$$\mathbf{F}_{\text{skip}} = \mathbf{F}_{\text{graph}} + \text{Conv}_5(\mathbf{U}_3). \tag{12}$$

Batch normalization is applied:

$$\mathbf{F}_{\text{bn}} = \text{BatchNorm}(\mathbf{F}_{\text{skip}}). \tag{13}$$

Finally, the output is summed with the identity mapping of $\mathbf{I}$ to produce the final feature map $\mathbf{O} \in \mathbb{R}^{W \times H \times C_{\text{in}}}$:

$$\mathbf{O} = \mathbf{F}_{\text{bn}} + \text{Conv}_6(\mathbf{I}). \tag{14}$$

Here, $\text{Conv}_6$ is a $1 \times 1$ convolution applied to the identity mapping $\mathbf{I}$, ensuring dimensional alignment with $\mathbf{F}_{\text{bn}}$ before summation. This preserves the residual learning behavior while allowing flexible channel depth.

#### Image nodes for retinal segmentation

In the RGNN block, each pixel in the OCTA image is treated as a graph node, with edges representing node-to-node distances, thereby enabling the modeling of relationships between retinal structures. The normalized affinity matrix $\mathbf{A}_{\text{norm}}$ captures these relationships, propagating information to enhance the segmentation of capillaries, veins, arteries, and the FAZ. This graph-based representation ensures that local vascular patterns and global dependencies are integrated for precise segmentation.

## Decoder

Following the relational feature extraction by the RGNN block, the decoder is responsible for reconstructing spatial resolution and refining anatomical boundaries across multiple retinal layers. This stage is critical for translating the graph-enhanced, semantically rich feature map into pixel-level vascular predictions. The decoder follows a symmetric structure, where each stage combines upsampling operations, residual refinement, and skip connections from the encoder. By progressively fusing multi-scale encoder outputs with the decoded feature maps, the decoder facilitates accurate segmentation of capillaries, arteries, veins, and FAZ. The use of residual units maintains information flow and mitigates vanishing gradients, while the skip pathways preserve spatial detail and vascular continuity, which are essential for delineating microvasculature. We write the encoder feature at stage $i$ as $\mathbf{f}_{e,i}$=$\mathbf{y}_i$. `ResidualA` and `ResidualB` are two residual conv blocks, and $\text{Up}_2$ is 2× nearest-neighbour upsampling. The decoder of RGNNNet reconstructs high-resolution spatial features, progressively restoring spatial granularity for segmenting retinal microvasculature, including capillaries, veins, arteries, and FAZ. Each decoder stage consists of upsampling followed by residual refinement and skip connections from encoder outputs. The following operations are performed in each decoding stage:

- **Stage 1:**
  The RGNN-enhanced feature map $f_{RGNN}$ is upsampled by a factor of 2, passed through Residual Units A and B, and concatenated with the encoder Stage 4 output $f_{e4}$:

$$f_{d1} = \text{Concat}(\text{ResidualB}(\text{ResidualA}(\text{Up}_2(f_{\text{RGNN}}))), f_{e4}) \tag{15}$$

  Since deeper encoder features encode global context, including FAZ geometry, fusing them here enhances the accurate delineation of FAZ boundaries.
- **Stage 2:**
  $f_{d1}$ is upsampled and passed through Residual Units, then concatenated with encoder Stage 3 output $f_{e3}$:

$$f_{d2} = \text{Concat}(\text{ResidualB}(\text{ResidualA}(\text{Up}_2(f_{d1}))), f_{e3}) \tag{16}$$

  This stage supports the precise segmentation of arteries by combining mid-level semantic features with refined spatial detail from earlier encoder stages.
- **Stage 3:**
  The refined features $f_{d2}$ are further upsampled, processed, and merged with the encoder Stage 2 output $f_{e2}$:

$$f_{d3} = \text{Concat}(\text{ResidualB}(\text{ResidualA}(\text{Up}_2(f_{d2}))), f_{e2}) \tag{17}$$

  This stage improves the detection of venous structures by preserving continuity in elongated vessels and enhancing contrast in lower-visibility regions.
- **Stage 4:**
  $f_{d3}$ is upsampled and combined with encoder Stage 1 output $f_{e1}$:

$$f_{d4} = \text{Concat}\left(\text{ResidualB}\left(\text{ResidualA}\left(\text{Up}_2\left(f_{d3}\right)\right)\right), f_{e1}\right) \quad (18)$$

This final skip connection restores fine-grained details crucial for capillary-level segmentation, preserving the microvascular morphology in high resolution.

– **Stage 5:**
The final decoder feature map $f_{d4}$ is upsampled and passed through a $1 \times 1$ convolution followed by sigmoid activation to yield the segmentation map:

$$f_{\text{final}} = \sigma\left(\text{Conv}_{1\times1}\left(\text{Up}_2\left(f_{d4}\right)\right)\right) \quad (19)$$

where $\sigma$ denotes the sigmoid function. The output map provides class-wise segmentation probabilities for capillaries, arteries, veins, and the FAZ.

For Output activation, we use one sigmoid head per class (arteries, veins, capillaries, FAZ), so each is learned as an independent binary task, which handles class imbalance better than Softmax. This decoder design ensures scale-adaptive refinement and precise anatomical reconstruction across retinal layers. Multi-scale skip connections combined with deep relational learning enable robust segmentation even in low-contrast or artifact-prone OCTA regions.

# Experiments and analysis

In this section, we introduce the input dataset utilized for training, provide additional experimental details, and present visual and quantitative comparisons of our proposed network.

## Dataset description & data augmentation

The proposed method was evaluated on the publicly available OCTA-500 dataset [26], which contains 500 subjects divided into two primary groups based on Field of View (FOV): OCTA 6 mm and OCTA 3 mm. The 6 mm subset includes 300 subjects (ID No. 10001–10300) with grayscale images at a resolution of $400 \times 400$, while the 3 mm subset consists of 200 subjects (ID No. 10301–10500), each having grayscale images with a resolution of $304 \times 304$. All images in the OCTA-500 dataset are provided in `.bmp` format. For each image in both the 3 mm and 6 mm subsets, the dataset includes corresponding ground truth segmentation masks for the vein, artery, capillary, and foveal avascular zone (FAZ). The images were acquired using a 70 kHz spectral-domain OCT system (RTVue-XR, Optovue, CA), and the dataset encompasses various retinal conditions such as retinal vein occlusion, choroidal neovascularization, age-related macular degeneration, and DR. Data augmentation is used to increase the variability of the training dataset and improve the model's robustness. Horizontal and vertical flips are applied with a probability of 0.5 to simulate different orientations of the images. Random brightness and contrast adjustments are made within a limited range of 0.1 to introduce variations in image appearance. The gamma transformation is applied with a 50 % probability to modify the intensity distribution across the images. Rotation up to 45° accounts for variations in image orientation. Elastic transformations and coarse dropout are used to simulate the effects of occlusion and deformation. Coarse dropout selectively removes randomly distributed patches from the images, simulating potential occlusions by varying the size and number of holes. For the OCTA-6 mm subset, the training set of 240 images was augmented to 960 images. For the OCTA-3 mm subset, the training set of 140 images was augmented to 560. This fourfold augmentation enhanced data diversity and model robustness. The OCTA-6M is not an independent dataset but the 6 mm FOV subset of OCTA-500, acquired on the same Optovue RTVue XR device and vice versa. In this work, we trained and evaluated models separately on the 3 mm and 6 mm subsets to ensure validation across different scan sizes within OCTA-500.

In addition to OCTA-500, we evaluated our models on the ROSE datasets. ROSE-0 [38] provides 117 FAZ-annotated OCTA images captured using an Optovue Avanti RTVue XR device with AngioVue software, each at a resolution of $304 \times 304$ pixels. Percentile stretch (2–98 %): To mitigate cross-dataset contrast shift, we optionally apply a training-free intensity remapping that linearly maps the 2nd and 98th image percentiles to 0 and 1, respectively. No retraining or parameter tuning is performed; the network and weights remain unchanged.

## Experimental settings

The proposed model was implemented in PyTorch and trained on a system equipped with an Intel Xeon W-3335 CPU at 3.34 GHz (16 cores), a 24 GB GDDR6X NVIDIA GeForce RTX 3090 GPU, and 128 GB of RAM. Preprocessing was carried out through a series of real-time augmentations designed to improve model generalization without increasing the dataset size. The pipeline included random horizontal and vertical flips, brightness and contrast adjustments, gamma correction, elastic deformation with $\alpha$=1.0, $\sigma$=50.0, and coarse dropout with a maximum of 4 holes of size $32 \times 32$. All transformations were implemented using the Albumentations library.

Training details: We train with Adam ($\beta_1$=0.9, $\beta_2$=0.999, $\epsilon$=$10^{-7}$) at an initial learning rate of $10^{-3}$. Each experiment uses a batch size of 32 and runs for up to 300 epochs with early stopping (patience 20). In validation, using two Residual Units per stage yielded consistently higher Dice and IoU than a single unit per stage across arteries, veins, capillaries, and FAZ, with no change to the training setup. The OCTA-500 dataset was used, with 240 images for training, 10 for validation, and 50 for testing from the 6 mm scans, and 140, 10, and 50 images, respectively, from the 3 mm scans. The data was split sequentially, following the same methodology described in [26]:

- **OCTA_6 mm**: The dataset is divided into 240 images for training (ID No. 10001–10240), 10 images for validation (ID No. 10241–10250), and 50 images for testing (ID No. 10251–10300).
- **OCTA_3 mm**: This subset contains 140 images for training (ID No. 10301–10440), 10 images for validation (ID No. 10441–10450), and 50 images for testing (ID No. 10451–10500).

### Loss function

A composite loss function combining Dice and Focal losses was employed to optimize segmentation accuracy. This combination effectively addresses class imbalance and enhances segmentation precision for delicate structures such as capillaries, arteries, veins, and the FAZ. Loss is computed per class head on sigmoid outputs and then averaged with class weights across arteries, veins, capillaries, and FAZ. The Dice loss $L_D$ compares the predicted probability $p_i$ with the ground-truth label $g_i$ for each pixel:

$$L_D = 1 - \frac{2\sum_i p_i\, g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \tag{20}$$

where $\epsilon$ is a small constant added for numerical stability. Here $p_i \in [0, 1]$ is the model prediction and $g_i \in \{0, 1\}$ is the ground-truth label.

We follow the soft Dice formulation used in prior work [39, 40].

The Focal loss $L_F$ focuses on challenging regions with low contrast, such as small capillaries, by down-weighting easily classified pixels:

$$L_F = -\beta_t \cdot (1 - q_t)^{\delta} \cdot \log(q_t) \tag{21}$$

where $q_t$ is the predicted probability for the target class, $\beta_t$ is a balancing factor, and $\delta$ controls the focus on harder examples. We use the Focal loss of Lin et al. [41] with $\beta_t$=0.25 and $\delta$=2 unless noted.

Hyperparameters and sensitivity: We follow standard Focal-loss settings for class-imbalanced segmentation [41] and write $\alpha$=$\beta_t$ and $\gamma$=$\delta$ for consistency with our notation. We examined small variations around these defaults during model selection and observed stable validation performance.

The total loss function is defined as:

$$L_{\text{total}} = L_D + L_F \tag{22}$$

In practice, $L_{\text{total}}$ is computed per head and then summed or averaged across the four heads.

This hybrid loss function enhances segmentation robustness by balancing pixel-wise accuracy with attention to challenging regions, thereby ensuring the reliable delineation of complex retinal structures in OCTA images. For convenience, the complete training configuration is summarized here: grayscale OCTA inputs, Adam optimizer (learning rate $1 \times 10^{-3}$, $\beta_1$=0.9, $\beta_2$=0.999, $\epsilon$=$10^{-7}$), batch size 32, maximum 300 epochs with early stopping (patience=20), encoder filters 16–96 with two Residual Units per stage, Dice + Focal loss ($\beta_t$=0.25, $\delta$=2), and independent sigmoid heads per class.

## Evaluation metrics

We assess the network's performance using five metrics: Dice ratio, Intersection over Union (IoU), Accuracy (ACC), Sensitivity (SE), and Specificity (SP). These metrics evaluate the quality of segmentation for arteries, veins, capillaries, and the FAZ in OCTA images.

The Dice ratio is computed as follows:

$$\text{Dice} = \frac{2 \cdot |A \cap B|}{|A| + |B|} \tag{23}$$

where $A$ represents the predicted segmentation and $B$ is the ground truth. The Dice ratio measures the similarity between segmented structures and their corresponding labels for arteries, veins, capillaries, and the FAZ, ensuring accurate delineation of vessels and avascular zones. The Intersection over Union (IoU) is defined as:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{24}$$

IoU quantifies the overlap between the predicted and ground truth segmentations. It is particularly useful for evaluating small structures, such as capillaries and the FAZ, where minor segmentation errors can significantly impact performance. The remaining metrics are computed as follows:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{25}$$

ACC represents the overall correctness of the segmentation by considering both correctly identified vessel and FAZ pixels, as well as correctly rejected background pixels. A high accuracy score indicates that the model effectively distinguishes between vascular structures and background regions.

$$\mathrm{SE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{26}$$

$$\mathrm{SP} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{27}$$

where:

- TP (True Positives): Correctly identified pixels belonging to arteries, veins, capillaries, or FAZ.
- TN (True Negatives): Correctly identified background (non-vessel or non-FAZ) pixels.
- FP (False Positives): Background pixels mistakenly classified as arteries, veins, capillaries, or FAZ.
- FN (False Negatives): Pixels belonging to arteries, veins, capillaries, or FAZ that were missed by the segmentation model.

For OCTA image analysis, these metrics help evaluate the precise segmentation of different vascular structures. SE is critical for detecting fine capillaries and avoiding under-segmentation of arteries and veins, while SP ensures the FAZ remains correctly delineated. The combination of these metrics provides a comprehensive assessment of the segmentation model's performance.

### Interpretability analysis

To examine the relational reasoning learned by the RGNN block, we visualize the symmetric, degree-normalized affinity matrix produced at the bottleneck layer in Figure 5. For a selected query pixel, the corresponding row of the affinity matrix is reshaped to the bottleneck grid and upsampled to the input resolution. Values are min–max normalized to [0, 1] and displayed with a colorbar. Query pixels are marked with white dots, and the predicted segmentation outline is overlaid for reference.

### Clinical validation

To evaluate the clinical robustness of the proposed framework, we conducted an additional blinded validation focused on the foveal avascular zone (FAZ), a clinically relevant OCTA biomarker. The FAZ module was tested under fixed model weights to prevent bias, and the reference FAZ masks supplied with the OCTA dataset served as the ground truth. Agreement between model predictions and reference annotations was assessed using Cohen's $\kappa$ statistic with 95 % confidence intervals computed via bootstrap resampling. This blinded design ensured that no manual adjustment or retraining influenced the outcome.

### Statistical testing

Statistical comparisons were performed using the Dice coefficient for each structure (capillary, artery, vein, and FAZ) and field of view (6 mm and 3 mm), as summarized in Table 2 (bottom). Overall differences among methods were assessed using the **Friedman test** with the **Iman–Davenport correction**, followed by **Nemenyi post-hoc comparisons** on average ranks to control the family-wise error rate ($\alpha$=0.05).

Additionally, treating the eight blocks as paired observations, we performed exact two-sided paired signed tests and exact Wilcoxon signed-rank tests vs. each baseline, and reported effect sizes (Vargha–Delaney $A_{12}$ and Cliff's $\delta$). Multiple pairwise p-values were controlled using Benjamini–Hochberg FDR ($q$=0.05).

## Results and discussions

### Objective assessment and comparative discussions

The proposed RGNNNet was evaluated against a comprehensive range of contemporary segmentation frameworks, including U-Net [12], Attention U-Net [13], UNet++ [14], UNet 3+ [15], CS-Net [21], AV-Net [20], TransUNet [16], and UNETR [17]. In addition to these well-established convolutional and transformer-based architectures, our evaluation also includes advanced graph-based methods such as Vision-GNN [30], G2ViT [31], and IPN-V2 [24]. These models were selected to ensure a fair and comprehensive benchmarking that reflects the evolution of OCTA segmentation techniques from standard U-Net derivatives to cutting-edge hybrid graph-transformer designs. Table 1 presents the comparative results for segmentation tasks involving capillaries, arteries, veins, and the FAZ, with metrics such as Dice, IoU, accuracy, sensitivity, and specificity used for evaluation. RGNNNet consistently demonstrated improved performance across these anatomical regions on both 3 mm and 6 mm subsets, confirming its robustness in multi-scale OCTA segmentation.

**Table 1:** Comparison of segmentation results for capillaries, arteries, veins, and FAZ against state-of-the-art methods. (Bold values indicate the best performance for each metric).

| Publications | 6 mm | | | | | 3 mm | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Dice | IoU | Accuracy | SE | SP | Dice | IoU | Accuracy | SE | SP |
| **Quantitative results of capillary segmentation across different methods** | | | | | | | | | | |
| Vision-GNN [30] | 87.52 | 76.34 | 90.59 | 87.21 | 90.98 | 89.41 | 79.02 | **91.82** | 89.74 | 91.26 |
| U-Net [12] | 84.63 | 73.31 | 88.06 | 84.87 | 90.07 | 86.96 | 76.98 | 88.37 | 90.62 | 86.78 |
| G2ViT [31] | 88.01 | 76.90 | **90.93** | 87.88 | 91.02 | 89.56 | 79.37 | 91.70 | 90.01 | 91.52 |
| UNet 3+ [15] | 87.11 | 77.17 | 89.95 | 88.42 | 90.88 | 87.56 | 77.84 | 89.62 | 90.15 | 89.31 |
| CS-Net [21] | 87.15 | 76.91 | 90.61 | 89.28 | 90.13 | 90.03 | 81.56 | 91.68 | 90.98 | 90.75 |
| Attention U-Net [13] | 87.46 | 77.58 | 90.27 | 89.11 | 90.98 | 89.64 | 81.25 | 91.43 | **91.53** | 91.42 |
| UNet++ [14] | 87.35 | 77.53 | 90.07 | 89.18 | 90.62 | 88.44 | 79.28 | 90.48 | 89.83 | 90.96 |
| UNETR [17] | 84.62 | 73.34 | 88.12 | 84.86 | 90.07 | 87.21 | 77.34 | 88.45 | 90.76 | 88.87 |
| AV-Net [20] | 87.43 | 77.67 | 90.31 | 89.03 | 90.85 | 89.52 | 81.04 | 91.36 | 91.51 | 91.18 |
| TransUNet [16] | 83.30 | 71.38 | 86.97 | 83.59 | 89.86 | 86.52 | 76.23 | 88.02 | 89.93 | 87.79 |
| IPN-V2 [24] | 87.64 | 77.65 | 87.82 | 87.12 | 88.01 | 90.06 | 81.52 | 90.32 | 89.83 | 89.90 |
| **Proposed work** | **88.33** | **78.12** | 90.67 | **89.91** | **92.47** | **90.60** | **81.93** | 91.72 | 90.62 | **92.65** |
| **Quantitative results of artery segmentation across different methods** | | | | | | | | | | |
| Vision-GNN [30] | 89.95 | 80.12 | 98.71 | 87.33 | 99.51 | 88.32 | 80.73 | 99.10 | 88.12 | 99.56 |
| U-Net [12] | 90.26 | 80.96 | 98.37 | 84.54 | 98.96 | 87.83 | 80.96 | 98.37 | 84.54 | 98.96 |
| G2ViT [31] | 90.04 | 80.89 | 98.74 | 87.90 | 99.57 | 88.48 | 81.04 | 99.13 | 88.33 | 99.59 |
| UNet 3+ [15] | 83.86 | 72.23 | 98.57 | 83.06 | 99.32 | 86.76 | 76.61 | 99.02 | 85.72 | 99.53 |
| CS-Net [21] | 79.87 | 67.23 | 98.35 | 78.56 | 99.26 | 82.14 | 69.97 | 98.75 | 79.78 | 99.47 |
| Attention U-Net [13] | 86.47 | 76.21 | 98.72 | 83.85 | 99.53 | 88.56 | 79.51 | **99.17** | 88.51 | 99.57 |
| UNet++ [14] | 84.53 | 73.18 | 98.67 | 83.10 | 99.35 | 87.46 | 77.73 | 99.09 | 87.95 | 99.47 |
| UNETR [17] | 85.79 | 75.11 | 98.56 | 83.26 | 99.42 | 86.72 | 76.54 | 98.97 | 85.63 | 99.48 |
| AV-Net [20] | 82.73 | 70.54 | 98.47 | 81.32 | 99.31 | 85.43 | 74.57 | 98.94 | 83.17 | 99.57 |
| TransUNet [16] | 84.66 | 73.42 | 98.52 | 83.28 | 99.28 | 85.54 | 74.71 | 98.87 | 83.20 | 99.52 |
| IPN-V2 [24] | 88.52 | 78.96 | 98.69 | 86.32 | 99.47 | 87.34 | 78.12 | 98.92 | 86.95 | 99.53 |
| **Proposed work** | **90.37** | **81.35** | **98.78** | **88.54** | **99.59** | **88.61** | **81.17** | 99.15 | **88.98** | **99.61** |
| **Quantitative results of vein segmentation across different methods** | | | | | | | | | | |
| Vision-GNN [30] | 89.33 | 81.14 | 99.22 | 85.76 | 99.64 | 88.06 | 80.94 | 99.20 | 84.61 | 99.67 |
| U-Net [12] | 85.11 | 74.08 | 98.53 | 86.74 | 99.15 | 84.81 | 73.66 | 99.13 | 83.51 | 99.58 |
| G2ViT [31] | 89.85 | 82.02 | 99.24 | 86.08 | 99.66 | 88.34 | 81.42 | 99.22 | **84.90** | 99.69 |
| UNet 3+ [15] | 83.47 | 71.64 | 98.45 | 83.14 | 99.17 | 85.36 | 74.45 | 99.13 | 84.08 | 99.61 |
| CS-Net [21] | 80.47 | 67.79 | 98.04 | 82.45 | 98.85 | 80.93 | 68.23 | 98.94 | 78.32 | 99.58 |
| Attention U-Net [13] | 86.41 | 76.12 | 98.73 | 85.07 | 99.42 | 85.67 | 74.96 | 99.17 | 84.76 | 99.58 |
| UNet++ [14] | 84.23 | 73.78 | 98.56 | 84.12 | 99.28 | 86.33 | 75.96 | 99.23 | 84.23 | 99.67 |
| UNETR [17] | 84.23 | 72.71 | 98.54 | 84.01 | 99.22 | 85.20 | 74.22 | 99.18 | 84.02 | 99.57 |
| AV-Net [20] | 83.59 | 71.79 | 98.48 | 81.37 | 99.34 | 70.63 | 54.57 | 98.25 | 71.89 | 99.05 |
| TransUNet [16] | 83.79 | 72.06 | 98.46 | 83.45 | 99.17 | 84.26 | 72.80 | 99.13 | 83.96 | 99.51 |
| IPN-V2 [24] | 88.22 | 80.63 | 99.15 | 85.92 | 99.60 | 87.25 | 80.48 | 99.19 | 84.32 | 99.66 |
| **Proposed work** | **90.13** | **82.41** | **99.27** | **86.91** | **99.69** | **88.67** | **82.41** | **99.25** | 84.83 | **99.70** |
| **Quantitative results of FAZ segmentation across different methods** | | | | | | | | | | |
| Vision-GNN [30] | 95.67 | 89.03 | 99.79 | 94.24 | 99.92 | 97.88 | 95.34 | 99.82 | 96.94 | 99.93 |
| U-Net [12] | 88.56 | 79.48 | 99.73 | 87.56 | 99.87 | 96.13 | 92.56 | 99.68 | 96.08 | 99.81 |
| G2ViT [31] | 96.12 | 89.71 | 99.80 | 94.86 | **99.93** | 97.96 | 95.70 | 99.83 | 97.18 | **99.94** |
| UNet 3+ [15] | 87.56 | 77.88 | 99.72 | 83.06 | 99.92 | 96.32 | 92.94 | 99.67 | 96.15 | 99.83 |
| CS-Net [21] | 86.15 | 78.04 | 99.67 | 85.84 | 99.88 | 95.47 | 91.75 | 99.64 | 95.32 | 99.83 |
| Attention U-Net [13] | 88.66 | 79.63 | 99.73 | 87.21 | 99.92 | 97.45 | 94.97 | 99.78 | 96.45 | 99.91 |
| UNet++ [14] | 81.32 | 68.50 | 99.62 | 77.78 | 99.86 | 96.30 | 92.87 | 99.64 | 96.38 | 99.76 |
| UNETR [17] | 89.72 | 81.35 | 99.77 | 88.75 | 99.87 | 96.43 | 93.14 | 99.64 | 96.37 | 99.85 |
| AV-Net [20] | 86.12 | 75.63 | 99.67 | 87.84 | 99.83 | 95.76 | 91.86 | 99.61 | 95.47 | 99.78 |

**Table 1:** (continued)

| Quantitative results of capillary segmentation across different methods | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 6 mm | | | | | 3 mm | | | | |
| **Publications** | **Dice** | **IoU** | **Accuracy** | **SE** | **SP** | **Dice** | **IoU** | **Accuracy** | **SE** | **SP** |
| TransUNet [16] | 88.67 | 79.66 | 99.72 | 87.62 | 99.88 | 95.54 | 91.47 | 99.62 | 95.38 | 99.83 |
| IPN-V2 [24] | 93.25 | 88.93 | 99.78 | 93.17 | 99.91 | 97.56 | 95.02 | 99.80 | 96.72 | 99.92 |
| **Proposed work** | **96.78** | **91.56** | **99.81** | **95.78** | **99.93** | **98.02** | **96.31** | **99.84** | **97.51** | **99.94** |

Attention U-Net showed slightly higher artery segmentation accuracy in the 3 mm case. Notably, G2ViT achieved a slightly higher accuracy for capillary segmentation on the 6 mm, while Vision-GNN also recorded marginally superior accuracy on the 3 mm compared to RGNNNet. However, across the majority of metrics, such as Dice and IoU, particularly crucial for structural precision, RGNNNet consistently demonstrated superior performance, underscoring its reliability in complex vascular delineation [30, 31].

Although our experiments center on diabetic retinopathy, the formulation is agnostic to disease. Separate sigmoid heads for each structure, and message passing in the graph operates on generic vascular cues that transfer to age-related macular degeneration and retinal vein occlusion. In these settings, FAZ irregularity and nonperfusion are common, and the relational propagation stabilizes predictions in sparsely perfused regions. With a small amount of fine-tuning on disease-specific cohorts, boundary fidelity and capillary recall improve. This supports the use of our approach for OCTA screening and follow-up beyond DR.

The outcomes demonstrate how well RGNNNet performs on difficult segmentation tasks, such as those involving complex structures like capillaries and the FAZ, which are crucial for the early diagnosis of DR. Additionally, RGNNNet is good at segmenting veins and arteries, which allows for a thorough examination of the vascular alterations linked to DR. By detecting pathological alterations like ischemia and vascular dilatation, the capacity to differentiate between these structures improves clinical evaluation. This characteristic highlights the model's versatility and therapeutic potential, mainly when precise capillary and FAZ segmentation is accomplished. These outcomes are significantly improved by adding the RGNN block, which successfully captures local and global links. The higher performance across various segmentation tasks strongly suggests that RGNNNet strikes an ideal compromise between accuracy and computational feasibility, even if computational benchmarks for the RGNN block were not explicitly examined. The inclusion of U-Net and its variants is consistent with prior work using the OCTA-500 dataset, where models such as U-Net, UNet++, and IPN-V2 have been routinely used as benchmarking references [26]. Similarly, transformer-based methods, such as Transunet and UNETR, reflect current trends in global attention mechanisms for fine-grained vessel delineation. CS-Net and AV-Net were chosen due to their specialization in artery-vein classification, which aligns closely with our multi-class segmentation objective. By further incorporating Vision-GNN and G2ViT, which represent recent developments in graph-based contextual learning, the benchmarking setup covers diverse modeling philosophies ranging from purely convolutional to graph-augmented and transformer-guided networks.

This diverse benchmarking suite not only adheres to standards seen in related OCTA literature but also enables meaningful comparison across architectures of varying complexity. The inclusion of these models was therefore not arbitrary, but rather intended to reflect the breadth of current state-of-the-art techniques relevant to our problem domain. RGNNNet's superior performance across these established baselines further reinforces its novelty and clinical relevance, particularly in handling non-perfusion regions, overlapping vessel types, and challenging anatomical zones, such as the FAZ. Table 2 provides a comparative analysis of the computational complexity of various state-of-the-art models. The memory usage of the proposed RGNNNet is notably lower (0.32 GB) compared to others while achieving a significantly faster inference speed (11.25 ms) for processing 400 $\times$ 400 OCTA images, and it has the fewest parameters (0.83 million) on an RTX 3090 GPU. All benchmark models were evaluated under a unified experimental setup, using identical input resolution, hardware configuration, and inference environment. This consistent setup allowed us to accurately estimate resource consumption and

performance, ensuring a fair comparison across different model architectures. This highlights the model's efficiency and suitability for real-time clinical applications where computational resources and response time are critical. Although the dataset provides only Normal and DR labels, we observed that segmentation errors were concentrated in the most severe DR images. Scans with extensive non-perfusion and highly disorganized vasculature – features typical of advanced stages – were particularly challenging. In these cases, the graph-based module sometimes produced fragmented vessel maps or irregular FAZ boundaries, reflecting the difficulty of modeling extreme vascular topologies.

**Table 2:** Computational Complexity, External Generalization (ROSE–0), and Statistical Significance Testing. *Top:* Complexity on an RTX 3090 (400 × 400). *Middle:* FAZ generalization on ROSE–0 (best threshold). *Bottom:* Nemenyi average ranks across eight evaluation blocks (Capillary/Artery/Vein/FAZ ×6 mm/3 mm), computed from Dice (higher is better); lower *average rank* (↓) indicates better performance. Friedman $\chi^2(11)=63.06$, p<0.001; Iman–Davenport $F(11, 77)=17.70$, p<0.001; Nemenyi CD=5.97 at $\alpha=0.05$. Methods marked[a] are significantly worse than the proposed work. (Bold values indicate the best performance for each metric).

| Method | Memory, GB | Params, M | Speed, ms |
|---|---|---|---|
| CS-Net [21] | 1.0 | 8.4 | 144 |
| U-Net [12] | 2.0 | 5.5 | 238 |
| AV-Net [20] | 2.4 | 6.5 | 158 |
| Attention U-Net [13] | 2.8 | 5.6 | 197 |
| UNet 3+ [15] | 4.9 | 16.1 | 243 |
| UNet++ [14] | 5.3 | 12.1 | 255 |
| Vision-GNN [30] | 5.6 | 11.2 | 211 |
| G2ViT [31] | 6.8 | 24.7 | 340 |
| TransUNet [16] | 6.2 | 18.9 | 310 |
| UNETR [17] | 6.4 | 21.5 | 328 |
| IPN-V2 [24] | 7.3 | 7.6 | 427 |
| **Proposed work** | **0.32** | **0.83** | **11.25** |

| Experiment | Best threshold | Dice | IoU | Acc | Sens | Spec |
|---|---|---|---|---|---|---|
| ROSE-0 baseline | 0.01 | 0.000 | 0.000 | 0.962 | 0.000 | 1.000 |
| ROSE-0 percentile stretch (2–98 %) | 0.08 | **0.915** | **0.843** | **0.994** | **0.908** | **0.997** |

| Method | Avg. rank ↓ | Δ vs. proposed | Sig.[(a)] |
|---|---|---|---|
| **Proposed work** | **1.00** | 0.00 | |
| G2ViT | 2.63 | 1.63 | |
| Vision–GNN | 3.88 | 2.88 | |
| IPN–V2 | 4.13 | 3.13 | |
| Attention U–Net | 5.00 | 4.00 | |
| U–Net | 7.50 | 6.50 | [a] |
| UNet++ | 7.81 | 6.81 | [a] |
| UNETR | 7.94 | 6.94 | [a] |
| UNet 3+ | 8.75 | 7.75 | [a] |
| AV–Net | 9.63 | 8.63 | [a] |
| TransUNet | 9.75 | 8.75 | [a] |
| CS–Net | 10.00 | 9.00 | [a] |

Nemenyi CD=5.97 at $\alpha=0.05$; [a]significantly worse vs. Proposed.

### External validation on ROSE-0

Without any pre-processing, the FAZ model underperforms on ROSE-0 due to domain shift (Dice≈0). Applying the simple percentile stretch (2–98 %) restores vessel/FAZ contrast and yields **Dice=0.915, IoU=0.843, Acc=0.994, Sens=0.908, Spec=0.997** at the best threshold (Table 2, middle).

### Statistical testing

Across 12 methods and eight blocks, the Friedman test indicated significant overall differences ($\chi^2(11)=63.06$, p<0.001), corroborated by the Iman–Davenport correction ($F(11, 77)=17.70$, p<0.001). Nemenyi's critical difference was 5.97; RGNNNet had the best average rank (1.00) and was significantly better than U–Net, UNet++, UNETR, UNet 3+, AV–Net, TransUNet, and CS–Net (rank gaps > CD) as summarized in Table 2 (bottom). Complementary paired, block–level tests were significant vs. every baseline (exact sign p=0.0078; exact Wilcoxon p=0.0078; BH–FDR $q=0.05$), with consistently positive effect sizes ($A_{12}>0.5$).

## Visual analysis

The segmentation performance of the proposed RGNNNet was evaluated against several state-of-the-art models using retinal images from the OCTA-500 dataset. As shown in Figures 3 and 4, the segmentation results include comparisons with original images, ground truths (GT), and outputs from models such as Vision-GNN [30], U-Net [12], G2ViT [31], UNet 3+ [15], CS-Net [21], Attention U-Net [13], UNet++ [14], UNETR [17], AV-Net [20], TransUNet [16], and IPN-V2 [24]. Across 3 mm and 6 mm imaging fields, the suggested model consistently demonstrates greater precision in defining fine details. RGNNNet produces findings that nearly match ground truth annotations and accurately segments the FAZ. The ability to accurately delineate FAZ borders is particularly significant, as morphological changes in the FAZ are critical indicators of disease progression in DR. The precise segmentation of these regions supports the early detection of retinal ischemia and capillary non-perfusion, critical markers of DR. RGNNNet is particularly adept at capturing capillary, arterial, and venous structures, in addition to FAZ segmentation, which is crucial for examining vascular changes associated with DR. The model's clinical value is enhanced by its ability to distinguish between arteries and

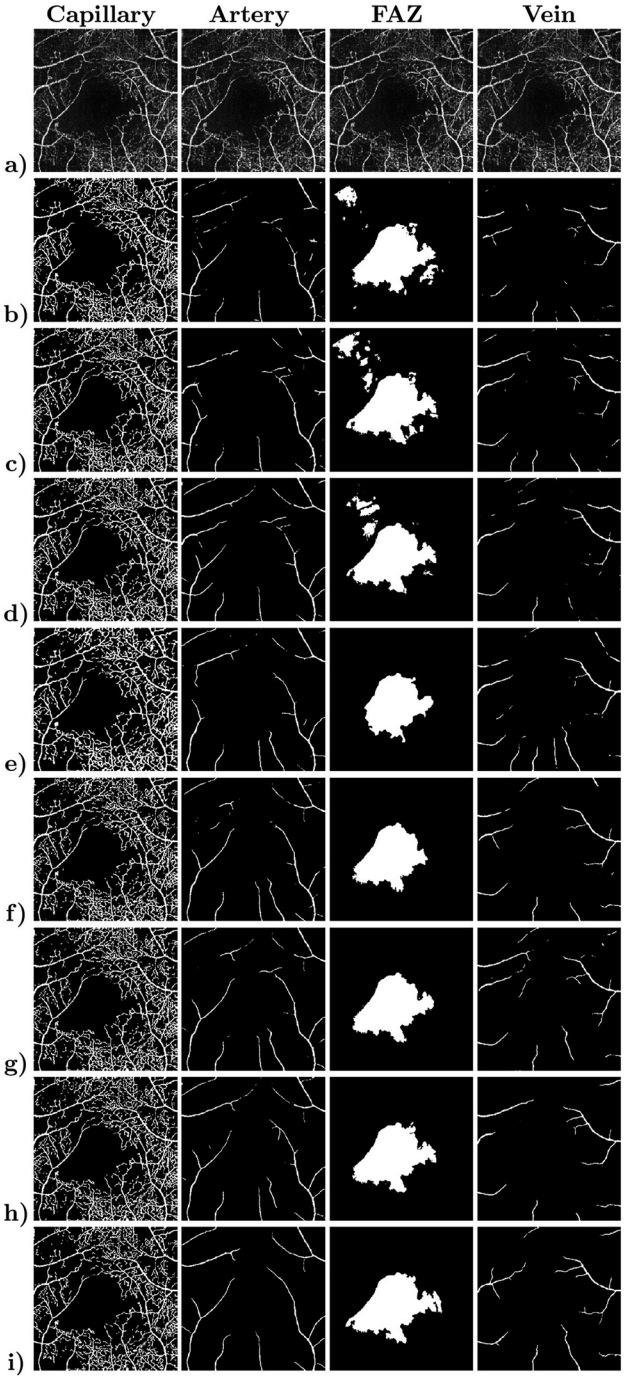



**Figure 3:** Segmentation results on 3 mm data across different models. (a) Original Images, (b) U-Net, (c) Attention U-Net, (d) U-Net++, (e) AV-Net, (f) TransUNet, (g) UNETR, (h) Proposed Method, (i) Ground Truth.

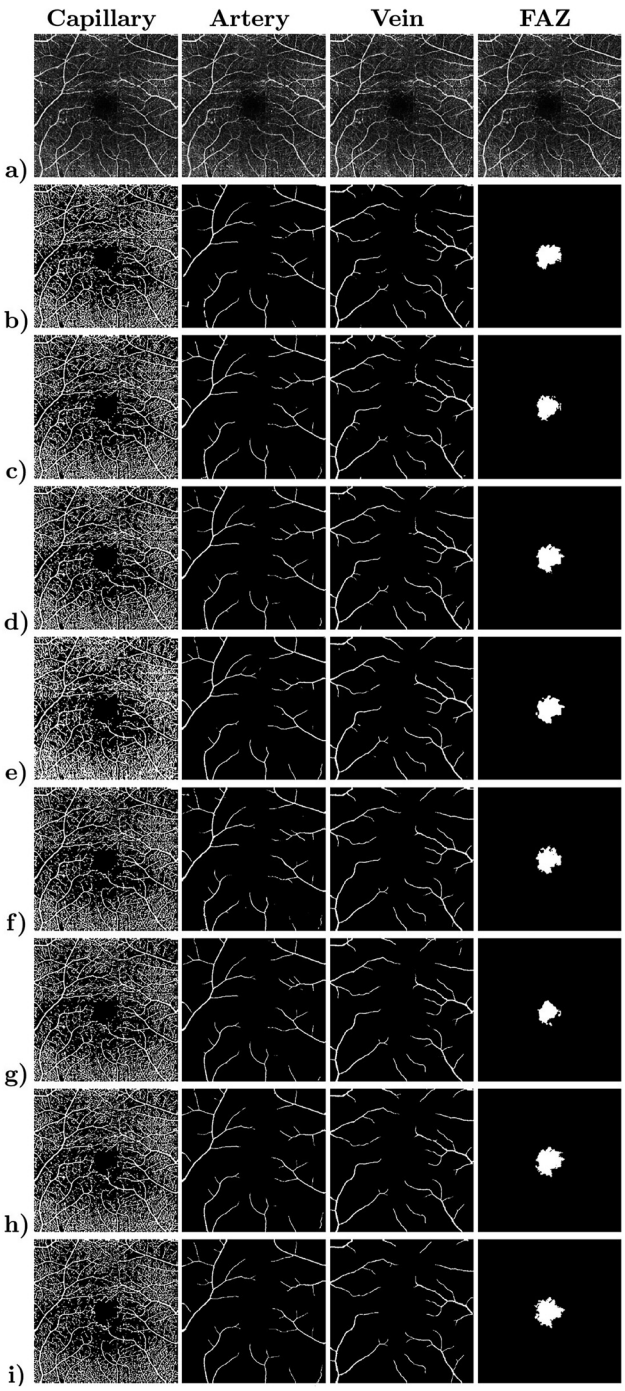



**Figure 4:** Segmentation results on 6 mm data across different models. (a) Original Images, (b) U-Net, (c) Attention U-Net, (d) U-Net++, (e) AV-Net, (f) TransUNet, (g) UNETR, (h) Proposed Method, (i) Ground Truth.

veins, which often undergo various pathological alterations as diseases progress.

Integrating the RGNN within RGNNNet further strengthens segmentation by effectively modeling local and global contexts. This capability enables the model to detect subtle vascular features critical for early-stage DR diagnosis and progression monitoring. The findings highlight RGNNNet's superior segmentation fidelity, which can

improve the evaluation of retinal illnesses. These developments give DR patients more precise retinal health evaluations and individualized treatment regimens.

## Ablation study

The proposed neural network architecture is subjected to an ablation study to determine the impact of various loss function variations on the segmentation outcomes. Capillaries, arteries, veins, and the FAZ are the four segmentation tasks used to assess the performance of loss functions, notably Dice and Focal, as well as their hybrid. Table 3 shows that using either Dice or Focal loss separately results in moderate segmentation performance. However, using the combination Dice + Focal, performance metrics are dramatically improved for all segmentation targets. The highest values of Dice and IoU are obtained from the Dice and Focal combination, especially for FAZ segmentation, attaining Dice values of 96.78 % and 98.02 %, and IoU values of 91.56 % and 96.31 % for OCTA-6 mm and OCTA-3 mm subsets, respectively. The consistently high performance across FAZ and retinal vasculature segmentation indicates that the Dice + Focal combination significantly enhances segmentation accuracy. Even though there are sporadic small increases in SP for the Dice loss, the overall ablation results support the need to include both Dice and Focal losses for improved segmentation outcomes.

To further assess the contribution of individual architectural components in RGNNNet, an ablation study was conducted by incrementally integrating three key modules: residual decoder blocks, adaptive skip refinement, and the RGNN block. As shown in Table 3, the baseline configuration consists solely of the encoder-decoder backbone with residual decoder blocks, without any skip or graph-based enhancements. Upon integrating adaptive skip refinement into the baseline, the segmentation performance improves across all vascular structures, particularly enhancing boundary precision and vessel continuity. Similarly, introducing the RGNN block alone to the baseline allows the network to capture broader contextual dependencies by modeling vascular relationships through graph representations. This results in higher Dice and IoU scores for all segmentation tasks. For instance, in the vein segmentation task (6 mm subset), the Dice score improves from 85.14 % (baseline) to 87.14 % with RGNN alone, and further rises to 90.13 % when both skip refinement and RGNN are integrated. The most significant performance gain, however, is achieved when both skip refinement and RGNN are combined with the residual decoder. This full configuration, marked as the proposed model, consistently achieves the highest accuracy across capillaries, arteries, veins, and the FAZ in both 6 mm and 3 mm subsets.

Beyond module-level ablations, we compare alternative edge-affinity definitions within the RGNN block: *cosine affinity* (cosine similarity between node embeddings) and a *Gaussian affinity* (RBF on feature distances), alongside a variant *without degree normalization*. As summarized in Table 3, these alternatives yield small but consistent drops compared to our degree-normalized RGNN, while the full configuration with *adaptive skip refinement + RGNN* remains best across vessels and FAZ. These results validate the effectiveness of each individual module and demonstrate the synergistic impact of their integration in enhancing structural coherence and fine vessel segmentation in OCTA images.

## Interpretability of RGNN

Figure 5 illustrates the relational cues captured by the RGNN module. For both a normal and a DR eye, a query placed at the FAZ center produces low affinity inside the avascular zone and high affinity along the parafoveal vascular ring, clarifying how the network delineates FAZ boundaries. A query placed on a vessel bifurcation yields strong affinity along the same branch and across the junction, demonstrating continuity modeling of the vascular topology. These visualizations provide intuitive evidence that dynamic graph construction supports both FAZ boundary recognition and vascular branch continuity, consistent with the quantitative performance gains.

## Clinical validation results

On 126 independent 6 × 6 mm OCTA scans, the FAZ segmentation achieved a Cohen's $\kappa$ of 0.951 (95 % CI: 0.941–0.960) against the reference masks. This shows excellent agreement and supports the reliability of our segmentation outputs. As our aim is structural segmentation, stage-level classification (NPDR/PDR) was not included due to a lack of labels.

**Table 3:** Ablation study showing the impact of Dice, Focal, and combined Dice + Focal losses on capillary, artery, vein, and FAZ segmentation across 6 mm and 3 mm OCTA scans, and ablation study on RGNNNet assessing the impact of key architectural components (6 mm and 3 mm).Affinity variants:"cosine affinity" computes edge weights via cosine similarity of node embeddings; "Gaussian affinity" uses an RBF over feature distances; "w/o normalization" omits degree normalization in the graph. (Bold values indicate the best performance for each metric).

| | 6 mm | | | | | 3 mm | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **Dice** | **IoU** | **Accuracy** | **SE** | **SP** | **Dice** | **IoU** | **Accuracy** | **SE** | **SP** |
| **Capillary segmentation** | | | | | | | | | | |
| Dice loss | 87.57 | 77.89 | 89.19 | 84.74 | 92.21 | 85.09 | 74.05 | 88.69 | 82.65 | 92.56 |
| Focal loss | 87.59 | 77.93 | 89.33 | 85.62 | 92.24 | 84.77 | 73.56 | 88.40 | 82.03 | 92.53 |
| Proposed (Dice + Focal) | **88.33** | **78.12** | **90.67** | **89.91** | **92.47** | **90.60** | **81.93** | **91.72** | **90.62** | **92.65** |
| **Artery segmentation** | | | | | | | | | | |
| Dice loss | 87.48 | 77.74 | 95.12 | 87.80 | 99.53 | 86.26 | 75.84 | 98.77 | 85.55 | 99.39 |
| Focal loss | 78.39 | 64.46 | 95.58 | 84.95 | 99.01 | 80.96 | 68.02 | 98.37 | 84.54 | 98.96 |
| Proposed (Dice + Focal) | **90.37** | **81.35** | **98.78** | **88.54** | **99.59** | **88.61** | **81.17** | **99.15** | **88.98** | **99.61** |
| **Vein segmentation** | | | | | | | | | | |
| Dice loss | 85.87 | 75.23 | 99.21 | 83.77 | 99.66 | 86.15 | 75.67 | 98.72 | **89.10** | 99.17 |
| Focal loss | 79.90 | 66.53 | 98.91 | 80.93 | 99.40 | 82.41 | 70.08 | 98.35 | 83.65 | 99.06 |
| Proposed (Dice + Focal) | **90.13** | **82.41** | **99.27** | **86.91** | **99.69** | **88.67** | **82.41** | **99.25** | 84.83 | **99.70** |
| **FAZ segmentation** | | | | | | | | | | |
| Dice loss | 94.09 | 89.35 | 99.77 | 93.34 | 99.74 | 95.47 | 91.33 | 94.26 | 96.71 | 99.93 |
| Focal loss | 94.31 | 89.87 | 99.74 | 93.47 | 99.82 | 89.92 | 81.69 | 98.77 | 90.66 | 99.87 |
| Proposed (Dice + Focal) | **96.78** | **91.56** | **99.81** | **95.78** | **99.93** | **98.02** | **96.31** | **99.84** | **97.51** | **99.94** |
| **Ablation results of capillary segmentation** | | | | | | | | | | |
| Baseline (residual decoder only) | 85.92 | 74.63 | 88.82 | 86.01 | 89.12 | 87.74 | 77.06 | 89.47 | 87.26 | 88.93 |
| Baseline + skip refinement | 87.22 | 76.82 | 89.94 | 88.32 | 91.06 | 89.23 | 79.61 | 90.85 | 89.38 | 90.11 |
| Baseline + RGNN only | 87.41 | 76.91 | 90.08 | 88.64 | 91.15 | 89.34 | 79.85 | 91.03 | 89.45 | 90.27 |
| Baseline + RGNN (w/o normalization) | 86.61 | 75.91 | 89.98 | 88.29 | 90.90 | 88.54 | 78.85 | 90.93 | 89.10 | 90.02 |
| Baseline + RGNN (cosine affinity) | 87.01 | 76.41 | 90.02 | 88.58 | 91.10 | 88.94 | 79.35 | 90.97 | 89.39 | 90.22 |
| Baseline + RGNN (Gaussian affinity) | 87.21 | 76.61 | 90.06 | 88.61 | 91.13 | 89.14 | 79.55 | 91.01 | 89.42 | 90.25 |
| Baseline + skip + RGNN (proposed) | **88.33** | **78.12** | **90.67** | **89.91** | **92.47** | **90.60** | **81.93** | **91.72** | **90.62** | **92.65** |
| **Ablation results of artery segmentation** | | | | | | | | | | |
| Baseline (residual decoder only) | 86.02 | 76.24 | 98.29 | 83.17 | 99.31 | 86.56 | 76.48 | 98.73 | 84.02 | 99.35 |
| Baseline + skip refinement | 88.33 | 78.74 | 98.62 | 85.66 | 99.45 | 87.34 | 78.21 | 98.91 | 86.22 | 99.56 |
| Baseline + RGNN only | 88.42 | 78.86 | 98.66 | 85.75 | 99.47 | 87.58 | 78.33 | 98.95 | 86.40 | 99.59 |
| Baseline + RGNN (w/o normalization) | 87.62 | 77.86 | 98.56 | 85.40 | 99.22 | 86.78 | 77.33 | 98.85 | 86.05 | 99.34 |
| Baseline + RGNN (cosine affinity) | 88.02 | 78.36 | 98.60 | 85.69 | 99.42 | 87.18 | 77.83 | 98.89 | 86.34 | 99.54 |
| Baseline + RGNN (Gaussian affinity) | 88.22 | 78.56 | 98.64 | 85.72 | 99.45 | 87.38 | 78.03 | 98.93 | 86.37 | 99.57 |
| Baseline + skip + RGNN (proposed) | **90.37** | **81.35** | **98.78** | **88.54** | **99.59** | **88.61** | **81.17** | **99.15** | **88.98** | **99.61** |
| **Ablation results of vein segmentation** | | | | | | | | | | |
| Baseline (residual decoder only) | 85.14 | 74.09 | 98.46 | 84.01 | 99.19 | 85.63 | 74.88 | 98.91 | 84.25 | 99.49 |
| Baseline + skip refinement | 87.05 | 76.80 | 98.78 | 85.82 | 99.47 | 86.21 | 76.31 | 99.16 | 85.17 | 99.63 |
| Baseline + RGNN only | 87.14 | 76.91 | 98.82 | 86.07 | 99.51 | 86.37 | 76.43 | 99.21 | **85.44** | 99.65 |
| Baseline + RGNN (w/o normalization) | 86.34 | 75.91 | 98.72 | 85.72 | 99.26 | 85.57 | 75.43 | 99.11 | 85.09 | 99.40 |
| Baseline + RGNN (cosine affinity) | 86.74 | 76.41 | 98.76 | 86.01 | 99.46 | 85.97 | 75.93 | 99.15 | 85.38 | 99.60 |
| Baseline + RGNN (Gaussian affinity) | 86.94 | 76.61 | 98.80 | 86.04 | 99.49 | 86.17 | 76.13 | 99.19 | 85.41 | 99.63 |
| Baseline + skip + RGNN (proposed) | **90.13** | **82.41** | **99.27** | **86.91** | **99.69** | **88.67** | **82.41** | **99.25** | 84.83 | **99.70** |

**Table 3:** (continued)

| Method | 6 mm | | | | | 3 mm | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Dice | IoU | Accuracy | SE | SP | Dice | IoU | Accuracy | SE | SP |
| **Ablation results of FAZ segmentation** | | | | | | | | | | |
| Baseline (residual decoder only) | 91.12 | 85.27 | 99.67 | 92.06 | 99.88 | 94.55 | 89.82 | 99.71 | 94.20 | 99.89 |
| Baseline + skip refinement | 93.84 | 88.42 | 99.75 | 94.61 | 99.91 | 96.33 | 93.04 | 99.79 | 95.88 | 99.92 |
| Baseline + RGNN only | 94.15 | 88.73 | 99.76 | 94.93 | 99.92 | 96.57 | 93.28 | 99.81 | 96.11 | 99.93 |
| Baseline + RGNN (w/o normalization) | 93.35 | 87.73 | 99.66 | 94.58 | 99.67 | 95.77 | 92.28 | 99.71 | 95.76 | 99.68 |
| Baseline + RGNN (cosine affinity) | 93.75 | 88.23 | 99.70 | 94.87 | 99.87 | 96.17 | 92.78 | 99.75 | 96.05 | 99.88 |
| Baseline + RGNN (Gaussian affinity) | 93.95 | 88.43 | 99.74 | 94.90 | 99.90 | 96.37 | 92.98 | 99.79 | 96.08 | 99.91 |
| Baseline + skip + RGNN (proposed) | **96.78** | **91.56** | **99.81** | **95.78** | **99.93** | **98.02** | **96.31** | **99.84** | **97.51** | **99.94** |

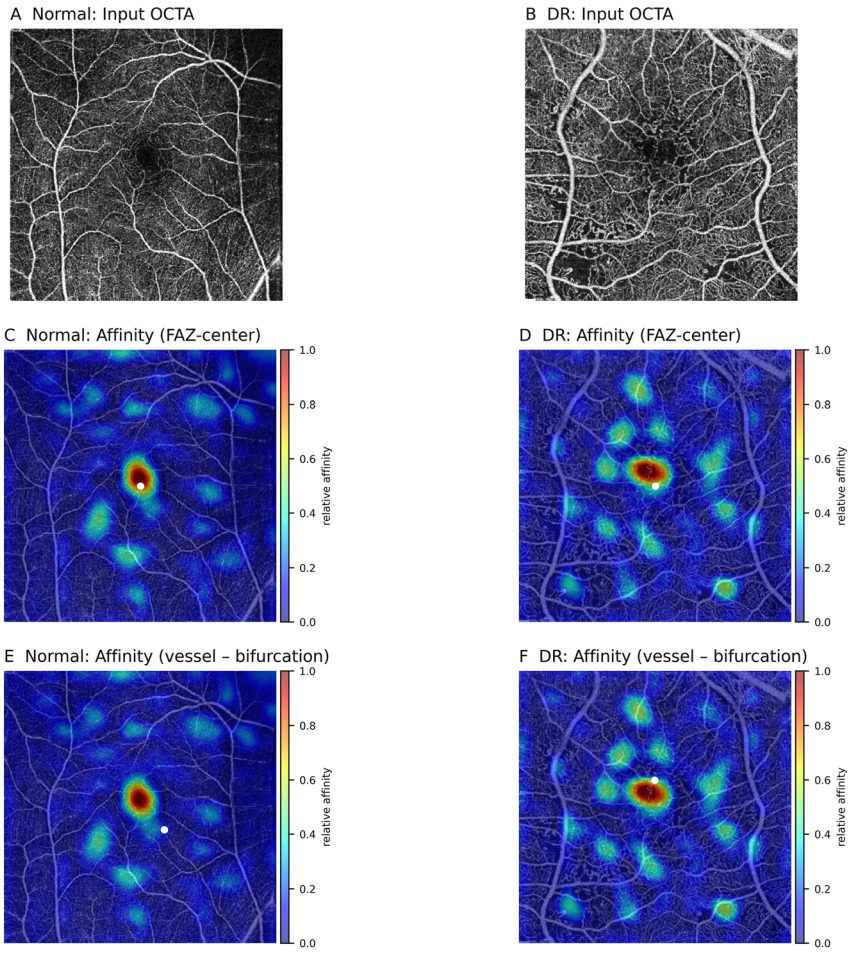



**Figure 5:** Interpretability of the RGNN module. For each case (Normal and DR), the top row shows the input OCTA, the middle row shows the affinity response to a query placed at the FAZ center, and the bottom row shows the affinity response to a query at a vessel bifurcation. White dots mark the query pixels. Color scale indicates symmetric, degree-normalized affinity values (min–max normalized to [0, 1]). Predicted mask outlines are overlaid. The FAZ-center query highlights the avascular boundary, while the vessel-bifurcation query demonstrates continuity modeling across branches.

## Conclusion and future work

This paper presented RGNNNet, a residual encoder–decoder framework enhanced with a Retinal Graph Neural Network (RGNN) block for precise segmentation of retinal vasculature and the foveal avascular zone (FAZ) in OCTA images. By integrating multi-scale feature extraction with relational modeling, the network achieves high-fidelity delineation of fine vascular structures. Experimental results on OCTA-500 (3 mm and 6 mm) confirm consistent improvements over

CNN, transformer, and graph-guided baselines. A training-free percentile stretch further enabled robust FAZ generalization to ROSE-0 without retraining. Rather than reiterating diagnostic claims, we emphasize that RGNNNet provides accurate structural masks that form a foundation for downstream biomarker extraction in diabetic retinopathy. The fundamental novelty of RGNNNet is its unified design: residual feature encoding combined with a dynamically constructed, symmetrically normalized graph module and adaptive skip refinement. This synergy allows the network to model both fine-grained retinal microvasculature and large-scale relational dependencies, a capability not present in prior CNN or transformer-based methods.

**Limitations.** First, due to computational cost, we did not perform exhaustive grid or Bayesian hyperparameter searches; encoder filter sizes (16–96), RGNN propagation iterations, and loss weights were empirically selected. Second, external validation was restricted to ROSE-0 because only this subset provides FAZ annotations compatible with our task; comprehensive vessel-type validation on ROSE-1 and ROSE-2 remains future work. Third, we focused on structural segmentation and did not attempt DR stage classification, since stage-level labels were unavailable. A limitation of the present study is the absence of stage-level annotations (NPDR/PDR), which prevents a quantitative, stage-specific failure analysis. Nevertheless, qualitative inspection indicates that the most difficult cases correspond to severe DR with markedly irregular vascular remodeling. Finally, current experiments used single-plexus projections; explicit multi-plexus modeling may further stabilize thin-vessel recovery.

**Outlook:** Future work will expand validation to multicenter cohorts with dense expert labels, incorporate cross-vendor calibration and domain adaptation for generalization across scanners, and integrate uncertainty estimation to support reliable confidence reporting. Extensions to multi-plexus inputs and lightweight post-processing (e.g., conditional random fields for enforcing exclusivity) may improve boundary fidelity. Future studies will also incorporate datasets with explicit stage labels and explore strategies such as uncertainty estimation and multi-plexus cues to better handle severe DR and other extreme vascular presentations. A prospective evaluation of clinical hardware and coupling RGNNNet masks with quantitative biomarker extraction (FAZ area, vessel density, and tortuosity) will advance translation toward early DR screening and monitoring.

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